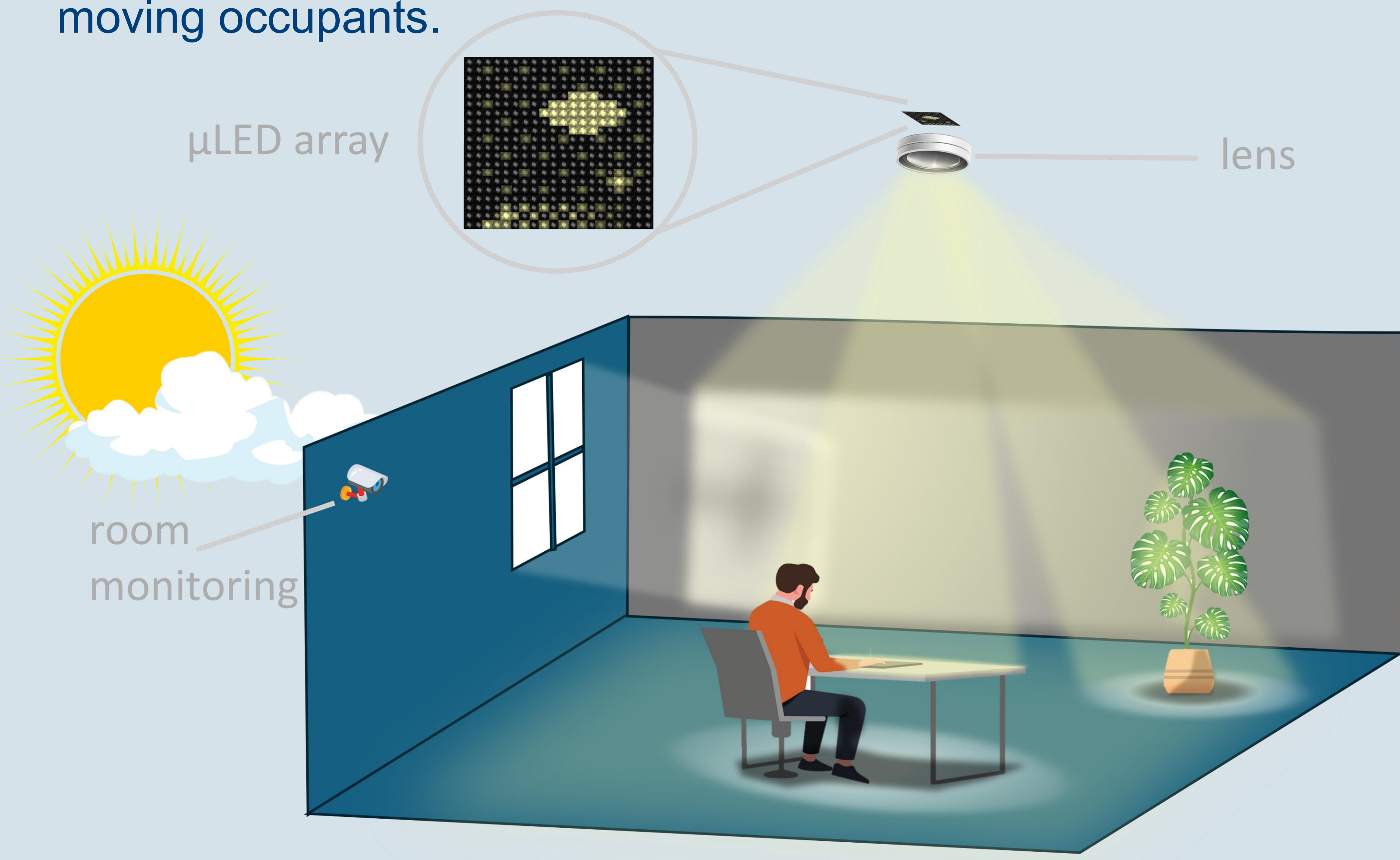




High-Fidelity Virtual Reality Framework for Human-centric Evaluation of Spatially Adaptive Lighting

Spatially adaptive lighting technology is here, but its integration into human-centric lighting design still lacks.

We envision an adaptive lighting system using high-resolution micro-LED arrays and lenses to dynamically shape light in response to monitored changing room conditions, including shifts in daylight spectrum caused by clouds, the time of day, and moving occupants.



What exists?

- ☒ μ LED hardware with projection optics
- ☒ Camera sensors for luminance and occupancy detection
- ☒ Proven dynamic light shaping

What lacks?

- ☐ Perception-based (spectral) control strategies
- ☐ Evaluation frameworks and standards
- ☐ Human-centric design methods

VR enables repeatable evaluation of lighting scenarios under identical emulated room and daylight conditions.

Challenges:



How can we quantify visual and non-visual stimuli?



How can we make VR physically accurate?



How do existing metrics perform in adaptive, non-uniform lighting?



How can we turn metric insights into practical control?

Tasks:



Build and validate a digital twin for real indoor environments



Calibrate VR for realistic eye-level exposure



Compare human response metrics and refine for adaptive lighting



Test context-aware control strategies against an optimal benchmark